

QC Methodology & Rules

Reference

How checks are defined and evaluated

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QC Methodology & Rules Reference

Castillo Planset QC — AI-Assisted QA/QC for Solar PV Plansets *Handoff reference for the QC Engineer and the Engineer of Record (PE)*

1. Purpose and Scope

This document is the authoritative reference for **how the Castillo Planset QC application decides what to check, how each check is evaluated, and how findings are classified**. It is written for the QC Engineer (who reviews the tool's findings each run) and the Engineer of Record (who is accountable for the sealed set and needs to understand exactly what the tool does and does not verify).

The QC engine combines three independent evaluation strategies over an uploaded planset PDF:

1. **Deterministic rules** — structural and text checks (sheet existence, keyword presence, drawing-index reconciliation, title-block presence) evaluated in `backend/app/analyzer.py` against the rule registry.
2. **Deterministic NEC electrical calculations** — first-principles engineering math in `backend/app/electrical_calcs.py`, run against specs auto-extracted from the planset (and/or supplied by the engineer).
3. **AI vision checks** — per-page and multi-page reasoning by the vision model, driven by structured prompts in `backend/app/gemini_analyzer.py`.

All findings from all three strategies are normalized into a single issue shape (status, severity, confidence, evidence, page, bounding box), de-duplicated, and rendered in the UI and the XLSX export.

The rule catalog itself lives in three YAML files:

File	Role
<code>backend/app/rules.yaml</code>	The active production rule registry — the checks described in this document.
<code>backend/app/rules_v4_draft.yaml</code>	The V4 draft rule set , ingested from the Castillo master QC checklist workbook. Used to enrich AI-finding descriptions and as the forward-looking, stage-gated rule model (not the default active registry).
<code>backend/app/stage_overrides.yaml</code>	The stage-gating overlay — the earliest design stage at which each rule/category should fire, plus explicitly disabled/duplicate rules.

STANDING NOTE — NEC edition. Every NEC table, article threshold, and correction factor encoded in the calculation engine and the AI validation prompts reflects the **NEC 2020 cycle**. Ampacities (310.16), temperature-correction factors (310.15), EGC/GEC sizing (250.122 / 250.66), working-space clearances (110.26), and PV rules (690.7 / 690.8 / 690.9) are all NEC-2020 values. When a jurisdiction has adopted a different code year, the engineer must confirm the affected values independently. A configurable, per-project NEC code-year selection is a planned enhancement and is **not** yet reflected in these hard-coded tables.

2. How a Rule Is Structured

Every rule in `rules.yaml` is a YAML mapping. The header of `rules.yaml` documents the schema; the fields are:

Field	Required	Meaning
<code>key</code>	yes	Unique identifier. Also the finding's <code>item_key</code> , used for de-duplication and traceability.
<code>category</code>	yes	Display grouping; must be one of the <code>category_order</code> values (Section 4).
<code>title</code>	yes	Short human-readable name shown on the finding card.
<code>description</code>	yes	Longer explanation of the intent — surfaced as the "why" on each finding.
<code>severity</code>	yes	Default severity: <code>low</code> \
<code>check_type</code>	yes	How the rule is evaluated (Section 3).
<code>nec_ref</code>	optional	NEC article reference(s), e.g. " <code>NEC 690.7</code> ", appended to evidence.
<code>handbook_ref</code>	optional	Checker's Handbook section reference.
<code>confidence</code>	optional	Default confidence, <code>0.0–1.0</code> .

Additional fields apply **per check type**:

Keyword checks

- **keywords** — a flat list of terms, **or** a list-of-lists (each inner group is an OR-set; the group counts as one match if any member is present).
- **min_matches** — minimum keyword matches required for a **Pass**.
- **search_scope** — where to search: **cover** \| **page_title** \| **sld** \| **dc** \| **tld** \| **relay** \| **electrical** \| **grounding** \| **site_plan** \| **pole_lineup**.
- **title_match** — sheet-title patterns that restrict the search to matching sheets.
- **exclude_title** — sheet-title patterns to exclude (e.g. SLD rules exclude **DC** and **AUX** sheets).

Sheet-existence checks

- **title_keywords** — title patterns matched against sheet titles to decide presence.

Electrical-calc checks

- **calc_function** — the name of the function in **electrical_calcs.py** to run.
- **inputs** — the **project_details** fields the calc consumes.

Representative rule examples

A **sheet-existence** rule:

```
- key: general_notes_page_exists
  category: General Notes
  title: General notes sheet exists
  severity: medium
  check_type: sheet_existence
  title_keywords: ["GENERAL NOTE", "GENERAL ELECTRICAL NOTE"]
```

A **grouped-keyword** rule:

```
- key: cover_sheet_project_info
  category: Cover Sheet
  title: Project Name, Address, Coordinates
  severity: high
  check_type: keyword
  search_scope: cover
  keywords:
    - ["PROJECT NAME", "PROJECT TITLE", "PROJECT:"]
    - ["SITE LOCATION", "SITE ADDRESS", "ADDRESS", "LOCATION"]
  min_matches: 2
  confidence: 0.95
```

An **electrical-calc** rule:

```
- key: sld_stringing
  category: AC Single Line Diagram
  title: "Stringing validation (V, I, P, MPPT, inputs, temperature)"
  severity: high
  check_type: electrical_calc
  calc_function: validate_stringing
```

```
nec_ref: "NEC 690.7"
confidence: 0.90
```

The V4 draft file (`rules_v4_draft.yaml`) adds two further fields per rule — `source` (the governing document, e.g. *"Approved module submittal"*) and `v4_status` (provenance tag: `V3`, `V4-FIX`, `V4-ENH`, `V4-NEW`) — which the AI-finding descriptions consume to show the engineer the source of intent.

3. Check Types and How Each Is Evaluated

`rules.yaml`'s header enumerates eight `check_type` values. Six are actively dispatched; the mechanics are:

3.1 `sheet_existence`

Matches the rule's `title_keywords` against the sheet titles found in the PDF.

- **Sheet present** → `Pass`, evidence `"Found: <sheet titles>"`, confidence fixed at **0.96**.
- **Listed in the drawing index but absent from the PDF** → `Fail`, severity forced to `high`, confidence **0.97**, anchored to page 1.
- **Neither present nor indexed** → the rule is out of scope and **skipped** (no finding emitted).

3.2 `keyword`

Resolves the `search_scope` (and optional `title_match` / `exclude_title`) to a set of pages, then case-insensitively searches the combined page text.

- Grouped keywords match one-per-group; flat keywords match individually.
- ``len(found) >= min_matches`` → `Pass` at the rule's `confidence`.
- **Otherwise** → `Needs Review` at `confidence × 0.85` (a keyword miss is treated as a prompt for human eyes, not an automatic failure).
- Matched terms are highlighted with bounding boxes on the page preview.

3.3 `drawing_index`

Reconciles the drawing index against the actual sheet set. This family covers index existence, all-indexed-sheets-present, no-extra-sheets, sheet-order-matches-index, and cover-sheet index consistency. Missing indexed sheets surface as high-severity fails; ordering and extra-sheet discrepancies are medium.

3.4 `title_block`

Verifies each page carries the required title-block fields (sheet number, sheet title). The AI title-block prompt enforces a strict **label-vs-value** discipline: printed template labels (`PROJECT NAME:`, `DATE:`) are

never treated as defects, and a field is only marked missing when its value cell is genuinely blank or shows a placeholder (XXXX, TBD, ---).

3.5 gemini_vision

Dispatches the page (rendered once to PNG and cached per page/zoom) to the vision model with a category-specific structured prompt. The model returns a JSON array of findings with **status/found**, **value**, **evidence**, **severity**, and location hints. This is the workhorse for anything requiring drawing comprehension — logos, maps, section views, equipment ratings, breaker phasing, winding configurations, clearances, consistency, and so on. Failed or unavailable AI calls degrade gracefully to **Needs Review**; the deterministic checks still stand.

Vision findings that come back without a usable location trigger a second-pass **bbox-rescue** call (deep model) that returns a bounding box for where the value is — or where it *should be* if the item is missing.

3.6 electrical_calc

Runs the named **calc_function** from **electrical_calcs.py** against the merged spec dictionary (auto-extracted planset specs overlaid with engineer-entered **project_details** and supporting-doc specs). See Section 5. A calc that cannot run because its inputs were not extracted is **not** reported as a defect: if it returns **Needs Review** with evidence beginning "Missing ", the status is flipped to **Deferred** ("not run — missing inputs") so it neither pollutes the review queue nor counts against precision.

3.7 manual

Items that require human judgment or external documents the tool cannot yet ingest (e.g. *compare poles with impact study*, *review submittal status*, *PVSyst summary*). These are carried in the registry at low default confidence (~0.50) as checklist reminders for the QC Engineer rather than automatically adjudicated findings.

3.8 regex

Reserved in the schema for pattern-based text checks; the active registry expresses its text matching through the **keyword** family.

4. Sheet Categories Reviewed

Findings are grouped and ordered by the **category_order** list in **rules.yaml**. The production categories, in display order, are:

4. Drawing Index
5. Title Block
6. Cover Sheet

7. System Information Table
8. General Notes
9. Site Plan
10. Pole Line Up
11. Engineered Equipment List
12. AC Single Line Diagram
13. DC Line Diagram
14. Three Line Diagram
15. Relay and Inverter Settings
16. AUX SLD
17. Communication Diagram
18. Feeder Plan
19. Communication Feeder Plan
20. Equipment Area Feeder Plan
21. Electrical Sheet
22. Inverter Zone Map
23. Elevation Details
24. Trenching Details
25. Overall Site Grounding Plan
26. Grounding Diagram
27. CAB or Cable Hanger Details
28. PAD / Slab Details
29. Pole Details
30. Labels
31. PVSyst Analysis Summary
32. Cross-Sheet Consistency
33. Supporting Document Consistency

The V4 draft registry ([rules_v4_draft.yaml](#)) organizes the same body of work around the **Castillo sheet-number taxonomy** (AI Input Gate, BOD / Due Diligence, E-001, E-002, E-010, E-011, E-050, E-100, E-101/E-102, E-103/E-104, E-106, E-107, E-110, E-120, E-130, E-140, E-200, E-210, E-214-E-217, E-300, E-400, E-420-E-422, E-450, E-500-E-504, E-601, E-900, Cross-Sheet). This taxonomy is what the stage-gating overlay keys against (Section 6).

5. The Deterministic NEC Calculation Engine

`electrical_calcs.py` is an **independent** engineering-math engine. It does not rely on the AI: it either validates values already extracted from the planset or recomputes them from first principles using the merged spec dictionary. Every function returns a `CalcResult` with `status` (`Pass` \ `Fail` \ `Needs Review`), `evidence`, `severity`, `confidence`, and a `computed` dictionary that captures the intermediate math for the finding card.

5.1 NEC reference tables encoded

Table / constant	NEC basis	Contents
NEC_310_16_CU_75C	310.16, 75 °C Cu	Ampacities #14–2000 kcmil (THWN-2/XHHW terminations)
NEC_310_16_CU_90C	310.16, 90 °C Cu	Ampacities #14–2000 kcmil (PV wire, USE-2)
NEC_310_16_AL_75C	310.16, 75 °C Al	Aluminum ampacities
PV_WIRE_90C	310.15(C)(1)	Free-air single-conductor PV-wire ampacities
NEC_250_122_EGC	250.122	Minimum EGC size vs OCPD rating
NEC_250_66_GEC_CU / _AL	250.66	GEC size vs largest ungrounded supply conductor
MV_15KV_DIRECT_BURIED_CU	310.60	15 kV-class direct-buried Cu ampacities
TRANSFORMER_BIL	transformer BIL by voltage class	15 kV→95, 25→125, 34.5→150, 46→200, 69→350 kV
NEC_110_26_CLEARANCES	110.26(A)(1)	Working-space clearances, Conditions 1/2/3, by voltage-to-ground
TEMP_CORRECTION_75C / _90C	310.15(C)(1)	Ambient temperature-correction factors (30 °C base)
CONDUIT_FILL_PCT	Chapter 9, Table 1	1 conductor 53 %, 2 conductors 31 %, 3+ conductors 40 %

5.2 Validation functions

The dispatcher `CALC_FUNCTIONS` maps each `calc_function` name to a function; `run_calc(name, project_details)` executes it. The functions and their engineering basis:

Function	Purpose	Key NEC basis / formula
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<code>validate_string_voc_cold</code>	String cold-temperature Voc $\leq$ inverter max DC input	690.7 ; Voc corrected to design low temp, xN modules
<code>validate_string_vmp_cold</code>	String cold Vmp $\leq$ inverter MPPT upper bound (startup clip check)	temp-corrected Vmp vs MPPT high
<code>validate_string_vmp_hot</code>	String hot Vmp $\geq$ inverter MPPT lower bound (hot off-peak check)	temp-corrected Vmp vs MPPT low
<code>validate_stringing</code>	Deprecated combined stringing check; retained for backward compatibility with legacy rules	690.7 (superseded by the three split functions above)
<code>validate_fuse_sizing</code>	String/DCB fuse $\geq 1.56 \times I_{sc}$, rounded to standard fuse size	690.9 ($1.56 = 1.25 \times 1.25$)
<code>validate_transformer</code>	Total XFMR kVA $\geq$ total inverter output; BIL per voltage class; Z% sanity	kVA adequacy, TRANSFORMER_BIL , ~5.75 % Z for 750–2500 kVA
<code>validate_dc_ampacity</code>	DC conductor ampacity after temp/fill derates	690.8 / 310.15 ; $I_{sc} \times 1.25$, $\div$ temp factor, sized from PV_WIRE_90C
<code>validate_ac_ampacity</code>	AC FLA, OCPD ($\times 1.25$ continuous), min conductor, parallel-set logic above 500 A	310.16 ; FLA = $kVA \cdot 1000 / (V \cdot \sqrt{3})$
<code>validate_mv_ampacity</code>	MV FLA, 1.25x design current, min MV cable	310.60 ; MV_15KV_DIRECT_BURIED_CU
<code>validate_egc_sizing</code>	Equipment grounding conductor from OCPD rating	250.122 ; NEC_250_122_EGC
<code>validate_gec_sizing</code>	Grounding electrode conductor from supply-conductor size	250.66 ; NEC_250_66_GEC_CU
<code>validate_voltage_drop</code>	VD presence/reasonableness (≤ 3 % total); requires cable lengths for exact calc	210.19(A) IN 4, 215.2(A)(4) IN 2 — returns Needs Review
<code>validate_conduit_fill</code>	Conduit fill < 40 % for 3+ conductors	Chapter 9, Table 1 — guidance Needs Review
<code>validate_nec_clearances</code>	Working-space clearances from system voltage-to-ground	110.26, Table 110.26(A)(1)
<code>validate_module_count</code>	Module Qty = String Size $\times$ String Quantity	exact arrangement math
<code>validate_total_dc</code>	Total DC $\approx$ Module Qty $\times$ STC W /	± 2 % binning tolerance; graduated

	1000	NR/Fail bands
<code>validate_total_ac</code>	Total AC = Inverter Qty × per-inverter kVA	exact; PF interpretation chosen to match stated total
<code>validate_dc_ac_ratio</code>	DC/AC ratio within 1.10–1.50 and matches stated	ratio math

Graduated tolerance behavior (important for the reviewer). The system-total checks (`validate_total_dc`, `validate_total_ac`) deliberately avoid false-positive failures: small deltas pass, mid-range deltas (up to ~10 %) and extreme deltas (> 50 %, almost always one wrong extracted input) are demoted to **Needs Review** with an extraction-quality hint, and only clearly-in-range designer-math errors are reported as **Fail**. The AC check additionally auto-selects unity-PF vs 0.9-PF interpretation to avoid flagging modern unity-PF inverter fleets.

Where the inputs come from. Before the calcs run, the pipeline auto-extracts specs from the planset pages (cover sheet, SLD, datasheets), then overlays engineer-entered `project_details` and any uploaded supporting-doc specs (BOD, CESIR, PVSyst, datasheets). User-entered values override auto-extracted ones. Provenance for every value is retained so the cross-sheet comparator (Section 7) can show where each number came from.

6. Statuses, Severities, and Confidence

6.1 Statuses and the UI color convention

Status	Meaning	Color
Pass	Check satisfied.	Green
Fail	Defect confirmed (missing required content, or math/rule violated).	Red
Needs Review	Ambiguous or unverifiable by the tool — human judgment required (keyword miss, unreadable value, VD/conduit-fill guidance, mid-range spec deltas).	Amber
Deferred	Not applicable or not run — gated out by design stage, or a calc whose inputs were not extracted.	Purple

6.2 Severity

Default severity is set per rule (**low** / **medium** / **high**). Calc functions and vision prompts can **escalate** dynamically — a confirmed math failure is forced to **high**, a merely-missing value stays **medium**, and cosmetic issues are **low**. Missing indexed sheets are always escalated to **high**.

6.3 What drives confidence

Confidence is a per-finding number in **0.0–1.0** reflecting how sure the tool is:

- **Deterministic structural checks** carry high fixed confidence — sheet-existence **Pass** = 0.96, missing-indexed-sheet **Fail** = 0.97.
- **Keyword checks** use the rule's declared **confidence** on **Pass**, and **`confidence x 0.85`** on **Needs Review** (a miss is less certain than a hit).
- **Electrical calcs** return their own confidence: ~0.88–0.92 for confirmed pass/fail with full inputs, dropping to ~0.40–0.75 when inputs are partial or the result is demoted to **Needs Review**.
- **AI vision findings** carry the confidence the model assigns to each finding, tempered by the rule's default; unavailable/failed calls degrade to **Needs Review**.
- **Manual reminders** sit at ~0.50 by design — they are checklist prompts, not adjudications.

The engineer should read confidence together with status: a low-confidence **Fail** or any **Needs Review** is an invitation to look, not a verdict.

7. Cross-Sheet Consistency and Supporting-Document Checks

7.1 Cross-sheet consistency

The **Cross-Sheet Consistency** category verifies that the same value agrees everywhere it appears across the set. Representative rules (all **gemini_vision**, severity **high**):

- **consistency_inverter_specs** — inverter make/model/kVA across SLD, 3LD, equipment list, site plan, DC diagram.
- **consistency_transformer_specs** — transformer kVA/voltage/winding across SLD, 3LD, equipment list.
- **consistency_cable_sizes** — AC/DC/MV cable sizes between SLD and the electrical schedule.
- **consistency_pole_names** — pole names/numbers across SLD, 3LD, Pole Line Up, Pole Elevation.
- **consistency_poi_voltage** — POI voltage identical on every sheet.
- **consistency_system_totals** — total DC/AC size across cover, SLD, equipment list, electrical schedule.
- **consistency_ct_vt_ratios** — metering CT/VT ratios across SLD, 3LD, relay settings.
- **consistency_breaker_ocpd** — breaker/OCPD ratings across SLD, 3LD, electrical schedule.

Deliberate exclusion. The system-information table is replicated identically across every sheet by template, so cross-sheet consistency *of that table* is guaranteed by construction. The vision prompts are explicitly instructed **not** to emit "sysinfo differs between pages" findings — the only relevant cross-source check is comparing the sysinfo table against the equipment datasheet.

7.2 Supporting-document / tech-spec cross-checks

The **Supporting Document Consistency** category compares the planset against uploaded reference documents (BOD, CESIR / interconnection study, approved submittals, datasheets, PVSyst). Specs parsed from those documents feed the same merged dictionary the calcs and comparator use, tagged with a `supporting:<doc_type>(<filename>)` provenance label. This lets the tool flag, for example, a grounding-transformer configuration that does not match the CESIR, a module electrical value that does not match the approved datasheet, or a transformer nameplate that disagrees with the approved submittal. Placeholder/blank values (TBD, not shown, N/A) are filtered out before comparison.

The V4 **AI Input Gate** and **BOD / Due Diligence** categories formalize this: before design checks run, they verify the governing inputs (approved module/inverter/transformer submittals, DC combiner and switchgear submittals, CESIR with appendices, utility ESB standards, voltage-drop workbook, thermal ampacity study) are present and project-specific rather than carried over from another project.

8. Design Stages and Stage-Based Gating

Not every check applies at every design stage. `stage_overrides.yaml` overlays the rule registry with **minimum-stage** gating.

8.1 Stage order

Stages are ordered:

```
30 < 60 < 90 < IFC < AsBuilt
```

When the engineer selects stage **X** for a run:

- Rules whose `min_stage` $\leq$ X fire normally.
- Rules whose `min_stage` $>$ X are **Deferred** (emitted as status **Deferred** with an "N/A at this stage" note) **unless** the corresponding sheet is actually present in the PDF, in which case the V4 engine's **lenient bypass** evaluates them anyway. The lenient bypass is intended for category defaults only.

8.2 Category minimum stages

`category_min_stage` assigns the default stage for every rule in a category (keyed to the V4 sheet taxonomy):

Stage	Categories gated at this stage
30 % (always required)	AI Input Gate, BOD / Due Diligence, Title Block, E-001, E-002, E-010, E-050, E-100, E-101/E-102, E-900
60 % (IFP / permit set)	E-120 (grounding plan), E-130 (lightning protection), E-140 (trenching), E-200 (electrical detail), E-300 (AC single-line), E-450 (PV harness layout), E-500–E-504 (schedules / feeder)
90 %	E-011 (reference/BOD), E-103/E-104 (AC details), E-110 (stringing plan), E-210 (electrical detail sub-family), E-400 (AC one-line)
IFC only	E-106, E-107, E-214–E-217 (structural/mechanical interface), E-420–E-422 (arc flash + protection coordination), E-601 (transmittal), Cross-Sheet (needs the full set to cross-reference)

8.3 Rule-level overrides and authoritative gating

`rule_overrides` lets individual outliers override their category default (keyed by `rule.key`, winning over `category_min_stage`). Example: the E-050 equipment-list rule `v4_e_050_no_tbd_at_ifc` is pushed to **IFC**, because TBD submittal entries are acceptable at 30/60/90 % and only firm up at IFC.

Overrides are authoritative. For any rule listed in `rule_overrides`, the lenient “fire anyway when the sheet is in the PDF” bypass is **disabled** — the rule was intentionally gated to a later stage and will not fire early just because the sheet happens to be present.

8.4 Disabled / duplicate-suppressed rules

`disabled_rules` lists rule keys that are **suppressed entirely** — never emitted as Pass/Fail/NR/Deferred and never dispatched to the vision model. This survives re-ingestion of V4 rules from Excel. It is used to resolve confirmed duplicates where the same check exists in a more appropriate category (for example, keeping the Cross-Sheet variant of an FLA or XFMR-primary-voltage match and disabling the sheet-specific copy, or keeping the NEC-cited conduit-fill variant for traceability). Each disabled entry carries an inline comment explaining the resolution. Notably, several NEC-code-year duplicates are collapsed to the single comprehensive cover-sheet vision check, and a bulk V3-vs-V4-ENH duplicate sweep disables the older V3 variants in favor of the more descriptive V4-ENH versions.

Maintenance guidance. Stage requirements and duplicate resolutions are meant to be tuned in `stage_overrides.yaml` — **not** in `rule_registry.py`. Editing the overlay keeps changes durable across re-ingesting the V4 checklist from the master Excel workbook.

9. Practical Reading Guide for the QC Engineer and PE

- **Green (Pass)** at high confidence is a verified item; still spot-check high-severity passes on a sealed set.
- **Red (Fail)** is a confirmed defect. For **electrical_calc** fails, the **computed** math is shown on the card — verify the extracted inputs are correct before acting, since a wrong extracted number can drive a real-looking fail.
- **Amber (Needs Review)** always means “the tool could not decide.” Keyword misses, unreadable drawing values, voltage-drop and conduit-fill items (which need cable lengths from the schedule), and mid-range spec deltas all land here by design.
- **Purple (Deferred)** means the check was intentionally not run — either gated out by the selected design stage or missing its calc inputs. It is not a defect and does not count against the set.
- **The NEC-2020 caveat applies to every electrical finding.** Confirm code-year applicability for the project’s jurisdiction until per-project code-year selection ships.
- **Manual-type items** are reminders the tool cannot yet automate (impact-study pole comparison, submittal status, PVSyst, DAS-vendor cross-checks). They remain the engineer’s responsibility.